TechVal AI — Laporan Penilaian Teknologi Ilmiah & Valuasi Ekonomi

Status: DISETUJUI PAKAR — REVISI TERVERIFIKASI | **Technology Code:** ITENAS-PAT-2026-001 | **Revision:** #1

DATA DEMONSTRASI SINTETIS — HANYA UNTUK VERIFIKASI SISTEM (TANPA KLAIM AKURASI)

Technology Title: Reaktor Membran Fotokatalitik Modular untuk Reklamasi Air Limbah Industri Tekstil

Domain: Environmental & Chemical Process Engineering | **IP Status:** granted

1. Ringkasan Eksekutif & Empat Luaran Independen

PRINSIP PEMISAHAN ILMIAH: Penilaian teknologi, kesiapan komersialisasi, valuasi ekonomi, dan kelengkapan bukti dipertahankan sebagai empat luaran terpisah. Skor penilaian dan TRL tidak pernah dikonversi menjadi nilai moneter ataupun parameter tingkat diskonto/pajak.

Luaran 1 — Penilaian Teknologi (Normalisasi 0–1 & Skala Asli TRL 1–9)	65.0% (0–1: 0.6500) | TRL 6 / 9 (Native NASA Scale)
Luaran 2 — Kesiapan Komersialisasi (Normalisasi 0–1)	65.0% (0–1: 0.6500)
Luaran 3 — Indikasi Valuasi Ekonomi (IDR)	Rp 666.060.121 (DCF Base Scenario (IDR))
Luaran 4 — Kelengkapan Bukti Tertimbang	100.0% (Observed: 10, Missing: 0, N/A: 0)

Provisional Status Notice: Aggregate assessment is labeled PROVISIONAL (Provisional equal weighting set is active.)

2. Provenans Analisis, Persetujuan & Manifes Reprodusibilitas

App Version	1.0.0
Requested / Actual Provider	fixture / fixture
Requested / Actual Model	fixture-deterministic-v1 / fixture-deterministic-v1
Prompt Hash (SHA-256)	9581a7d6d653e5de0b07c44d6c5c2131aefcf21125e4f707
Run Config Hash	a0fb65484f9fd4414391252681b9c02b6512fc47eb1b2a1e
Rubric & Weight Version	methodology_v1.0 | provisional_equal_v1
Document Hashes (SHA-256)	936f24bb:28fe943cfb33..., bc47a01e:196966f748c6...

3. Rubrik Penilaian Teknologi & Kesiapan Komersialisasi

Code	Dimension	Status	Raw / Norm	Weight (Renorm)	Rationale
COMM_COMPETITIVE_POSITION	commercialization_readiness	observed	4 / 0.75	0.20 (0.20)	Evaluasi COMM_COMPETITIVE_POSITION pada level 4 berdasarkan bukti dokumen terverifikasi pada page:1.
COMM_IP_LICENSING_READINESS	commercialization_readiness	observed	3 / 0.50	0.20 (0.20)	Evaluasi COMM_IP_LICENSING_READINESS pada level 3 berdasarkan bukti dokumen terverifikasi pada para:3.
COMM_MARKET_DEMAND	commercialization_readiness	observed	4 / 0.75	0.20 (0.20)	Evaluasi COMM_MARKET_DEMAND pada level 4 berdasarkan bukti dokumen terverifikasi pada page:1.
COMM_REGULATORY	commercialization_readiness	observed	3 / 0.50	0.20 (0.20)	Evaluasi COMM_REGULATORY pada level 3 berdasarkan bukti dokumen terverifikasi pada para:4.

COMM_SUPPLY_CHAIN	commercialization_readiness	observed	4 / 0.75	0.20 (0.20)	Evaluasi COMM_SUPPLY_CHAIN pada level 4 berdasarkan bukti dokumen terverifikasi pada page:3.
TECH_NOVELTY	technology_assessment	observed	3 / 0.50	0.20 (0.20)	Evaluasi TECH_NOVELTY pada level 3 berdasarkan bukti dokumen terverifikasi pada page:1.
TECH_PATENT_STRENGTH	technology_assessment	observed	4 / 0.75	0.20 (0.20)	Evaluasi TECH_PATENT_STRENGTH pada level 4 berdasarkan bukti dokumen terverifikasi pada page:1.
TECH_SCALABILITY	technology_assessment	observed	3 / 0.50	0.20 (0.20)	Evaluasi TECH_SCALABILITY pada level 3 berdasarkan bukti dokumen terverifikasi pada page:2.
TECH_SYSTEM_INTEGRATION	technology_assessment	observed	4 / 0.75	0.20 (0.20)	Evaluasi TECH_SYSTEM_INTEGRATION pada level 4 berdasarkan bukti dokumen terverifikasi pada page:2.
TECH_TRL_NATIVE	technology_assessment	observed	6 / Native	0.00 (0.00)	Evaluasi TECH_TRL_NATIVE pada level 6 berdasarkan bukti dokumen terverifikasi pada page:2.
TECH_VALIDATION	technology_assessment	observed	4 / 0.75	0.20 (0.20)	Evaluasi TECH_VALIDATION pada level 4 berdasarkan bukti dokumen terverifikasi pada page:2.

4. Bukti Dokumen Terverifikasi & Kutipan Sumber Asli

• **[TECH_NOVELTY]** Doc: 936f24bb | Locator: **page:1** | Status: **resolved**
Original Source Quotation: "Technical Architecture & Novelty (Page 1) The disclosed invention integrates a TiO2-graphene nanocomposite coating immobilized on a tubular ceramic ultrafiltration m Unlike conventional coagulation-flocculation processes that generate hazar"

• **[TECH_PATENT_STRENGTH]** Doc: 936f24bb | Locator: **page:1** | Status: **resolved**
Original Source Quotation: "SYNTHETIC DEMONSTRATION DOCUMENT · ITENAS RESEARCH PORTFOLIO Document Type: Patent Specification & Technical Disclosure Patent Number: IDP000089421 (Granted in Indonesia, PDKI) Technology Title: Modular Photocatalytic Membrane Reactor for I"

• **[TECH_SCALABILITY]** Doc: 936f24bb | Locator: **page:2** | Status: **resolved**
Original Source Quotation: "Pilot campaign conducted over 450 cumulative operating hours treating 15 m3/day of raw dyeing effluent"

• **[TECH_SYSTEM_INTEGRATION]** Doc: 936f24bb | Locator: **page:2** | Status: **resolved**
Original Source Quotation: "Engineering scaling parameters, transmembrane pressure profiles (1"

• **[TECH_TRL_NATIVE]** Doc: 936f24bb | Locator: **page:2** | Status: **resolved**
Original Source Quotation: "Technology Readiness Level (TRL) & Subsystem Validation (Page 2) The subsystem prototype has been demonstrated in a relevant industrial environment at TRL 6 exit criteria"

• **[TECH_VALIDATION]** Doc: 936f24bb | Locator: **page:2** | Status: **resolved**
Original Source Quotation: "Pilot campaign conducted over 450 cumulative operating hours treating 15 m3/day of raw dyeing effluent"

• **[COMM_COMPETITIVE_POSITION]** Doc: 936f24bb | Locator: **page:1** | Status: **resolved**
Original Source Quotation: "Technical Architecture & Novelty (Page 1) The disclosed invention integrates a TiO2-graphene nanocomposite coating immobilized on a tubular ceramic ultrafiltration m Unlike conventional coagulation-flocculation processes that generate hazar"

• **[COMM_IP_LICENSING_READINESS]** Doc: bc47a01e | Locator: **para:3** | Status: **resolved**
Original Source Quotation: "Initial pilot licensee letter of intent signed for 3 units"

• **[COMM_MARKET_DEMAND]** Doc: 936f24bb | Locator: **page:1** | Status: **resolved**
Original Source Quotation: "SYNTHETIC DEMONSTRATION DOCUMENT · ITENAS RESEARCH PORTFOLIO Document Type: Patent Specification & Technical Disclosure Patent Number: IDP000089421 (Granted in Indonesia, PDKI) Technology Title: Modular Photocatalytic Membrane Reactor for I"

• **[COMM_REGULATORY]** Doc: bc47a01e | Locator: **para:4** | Status: **resolved**
Original Source Quotation: "Regulatory & Standards Readiness: Effluent quality meets Ministry of Environment Regulation PermenLHK

No"

• **[COMM_SUPPLY_CHAIN]** Doc: 936f24bb | Locator: **page:3** | Status: **resolved**
Original Source Quotation: "Intellectual Property Claims & Manufacturing Readiness (Page 3) Independent Claim 1 covers the thermal-sol-gel deposition method bonding the catalytic layer onto alumina supports withou Domestic supply chain audit confirms 82% local compone"

5. Valuasi Ekonomi Deterministik (IDR) & Asumsi Keuangan

Scenario & Approach	Attribution / Risk	Indicative Value (IDR)	WIPO Method Locator
DCF Base Scenario (IDR) [dcf]	documented_attribution_factor (0.35) | in_discount_rate	**Rp 666.060.121**	WIPO IP Valuation Basics Guide, Section 3.3 (Income Method — Discounted Cash Flow), pp. 26-34
DCF Downside / Low Scenario (IDR) [dcf]	documented_attribution_factor (0.35) | in_discount_rate	**Rp 315.724.137**	WIPO IP Valuation Basics Guide, Section 3.3 (Income Method — Discounted Cash Flow), pp. 26-34
DCF Upside / High Scenario (IDR) [dcf]	documented_attribution_factor (0.35) | in_discount_rate	**Rp 1.057.868.593**	WIPO IP Valuation Basics Guide, Section 3.3 (Income Method — Discounted Cash Flow), pp. 26-34
Replacement Cost Approach (IDR) [cost]	direct_cash_flows (1.0) | in_discount_rate	**Rp 826.500.000**	WIPO IP Valuation Basics Guide, Section 3.1 (Cost Method — Reproduction & Replacement Cost), pp. 18-21
Relief-from-Royalty Approach (IDR) [relief_from_royalty]	direct_cash_flows (1.0) | in_discount_rate	**Rp 489.508.230**	WIPO IP Valuation Basics Guide, Section 3.3.2 (Relief-from-Royalty Method), pp. 31-34
Market Comparables Indication Range (IDR) [market_comparables]	direct_cash_flows (1.0) | in_discount_rate	**Rp 979.800.000**	WIPO IP Valuation Basics Guide, Section 3.2 (Market Method — Comparable Transactions), pp. 22-25

6. Konvensi Metodologi, Keterbatasan & Referensi Ilmiah

• Normalization Formula: 1–5 rubrics normalized via (score - 1) / 4; TRL retained on native 1–9 scale.
• AHP Convention: Consistency Ratio CR <= 0.10 enforced as a methodological convention (Saaty, 1980).
• Patent Indicators: Forward citations and family size treated strictly as contextual signals (Harhoff et al., 2003).
• Valuation Separation: Different valuation approaches (DCF, Cost, Comparables, Royalty) are presented separately and never averaged automatically.

• **[HARHOFF-2003]** Harhoff, D., Scherer, F. M., & Vopel, K. (2003). *Citations, family size, opposition and the value of patent rights*. Research Policy, 32(8), 1343-1363. [Bib: verified | Method: contextual_only | Locator: Sections 3-5, Tables 3-6, pp. 1348-1360]

• **[NASA-TRL-ESTO]** Mankins, J. C. / NASA Earth Science Technology Office (ESTO) (1995). *Technology Readiness Levels: A White Paper & NASA ESTO TRL Definitions and Exit Criteria*. NASA Office of Space Access and Technology / ESTO. [Bib: verified | Method: verified_with_locator | Locator: NASA TRL Scale Definitions (Levels 1-9) & Exit Criteria Table, Sections 1-3, pp. 1-6]

• **[SAATY-AHP-1980]** Saaty, T. L. (1980). *The Analytic Hierarchy Process: Planning, Priority Setting, Resource Allocation*. McGraw-Hill International Book Co., New York. [Bib: verified | Method: verified_with_locator | Locator: Chapter 1, Section 1.5 (Eigenvalue Priority Vector & Consistency Ratio), Table 1.5 (Random Index RI), pp. 17-24]

• **[SHROUT-FLEISS-1979]** Shrout, P. E., & Fleiss, J. L. (with Cohen, J., 1968) (1979). *Intraclass correlations: Uses in assessing rater reliability (and Weighted Kappa: Nominal scale agreement with provision for scaled disagreement)*. Psychological Bulletin, 86(2), 420-428 & 70(4), 213-220. [Bib: verified | Method: verified_with_locator | Locator: Shrout & Fleiss (1979) Formulas for Case 2 ICC(2,1) and Case 3 ICC(3,1), Table 1, pp. 422-425; Cohen (1968) Eq. 6-8, pp. 215-217]

• **[WIPO-VAL-2024]** World Intellectual Property Organization (WIPO) (2024). *Intellectual Property Valuation Basics for Technology Transfer Professionals*. WIPO Web Publications, Geneva. [Bib: verified | Method: verified_with_locator | Locator: Section 3.1 (Cost Method, pp. 18-21); Section 3.2 (Market Method, pp. 22-25); Section 3.3 (Income/DCF & Relief-from-Royalty, pp. 26-39)]